

Practical 8

system has RAM which can store four pages simultaneously to satisfy page

requests. Write a program to calculate percentage page hit and page fault if the

system uses Least recently used page replacement technique for the page

references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

```
GNU nano 6.2 lru.c *
#include <stdio.h>

int main()
{
    int frames = 3;
    int pages[12] = {1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5};
    int frame[3];
    int time[3];

    int i, j, k;
    int pageFault=0, pageHit=0;
    int counter=0;

    for(i=0; i<frames; i++)
        frame[i] = -1;

    for(i=0; i<12; i++)
    {
        int flag=0;

        for(j=0; j<frames; j++)
        {
            if(frame[j]==pages[i])
            {
                counter++;
                time[j]=counter;
                pageHit++;
                flag=1;
                break;
            }
        }

        if(flag==0)
        {
            int pos=0;
            int min=time[0];
```

```

tanuja@Tanuja: ~
GNU nano 6.2 lru.c *
        break;
    }
}

if(flag==0)
{
    int pos=0;
    int min=time[0];

    for(j=1;j<frames;j++)
    {
        if(time[j]<min)
        {
            min=time[j];
            pos=j;
        }
    }

    frame[pos]=pages[i];
    counter++;
    time[pos]=counter;
    pageFault++;
}

printf("Total Page Hits = %d\n",pageHit);
printf("Total Page Faults = %d\n",pageFault);

float hitPer = (pageHit/12.0)*100;
float faultPer = (pageFault/12.0)*100;

printf("Page Hit Percentage = %.2f%%\n",hitPer);
printf("Page Fault Percentage = %.2f%%\n",faultPer);

return 0;
}

```

Output:

```

This message is shown once a day. To disable it please create the
/home/tanuja/.hushlogin file.
tanuja@Tanuja:~$ nano lru.c
tanuja@Tanuja:~$ gcc lru.c
tanuja@Tanuja:~$ ./a.out
Total Page Hits = 0
Total Page Faults = 12
Page Hit Percentage = 0.00%
Page Fault Percentage = 100.00%
tanuja@Tanuja:~$ |

```